Prodigy Anesthesia Quick Review

Glaucoma

What is glaucoma?	Glaucoma is an increase in intraocular pressure that results in a decrease in blood flow to the optic nerve and it can progress to optic nerve damage and loss of vision. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 4.*
What are the two types of glaucoma?	Open-angle (chronic) glaucoma and closed-angle (acute) glaucoma. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 4.*
What glaucoma medication can prolong the effects of succinylcholine?	Echothiophate is used to treat glaucoma and irreversibly antagonizes cholinesterase. Because of this effect, it will prolong the duration of action of succinylcholine. The duration of paralysis usually does not exceed 20-30 minutes, however. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 763.*
What is the normal intraocular pressure?	The normal intraocular pressure is about 10 to 21 mmHg. Pressures above 22mmHg are considered intraocular hypertension. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1376.*
What are the two factors that determine intraocular pressure?	The two factors that determine intraocular pressure are the rate of formation of aqueous humor and the rate of drainage of the aqueous humor. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 4.*
What physiologic factors will increase intraocular pressure?	Factors that increase the intraocular pressure include: increased central venous pressure, increased arterial blood pressure, increased PaCO2 (hypoventilation), and hypoxia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 760.*
At what point during the day is intraocular pressure the highest?	Intraocular pressure is higher in the morning, presumably from venous congestion due to the eyes being closed. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1376.*

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What is the pathophysiologic difference between open-angle and closed-angle glaucoma?	Open-angle glaucoma is characterized by a patent anterior chamber angle and the drainage of aqueous humor is impaired by sclerosis of the trabecular tissue. In closed-angle glaucoma, the drainage of aqueous humor is impaired by a shallow anterior chamber and a narrow iridocorneal angle that results in the trabecular network being obstructed by the iris. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 4-5.*
What are the symptoms of acute angle-closure glaucoma?	Ocular pain, blurred vision, and fixed, mid-dilated pupils. If not treated promptly, permanent visual loss can occur due to damage to the optic nerve. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 6.*
What is the treatment for acute angle-closure glaucoma?	If reduction in intraocular pressure doesn't occur promptly with medical treatment, surgical iridotomy/iridectomy should be performed. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 5-6.*
What is the principal treatment for open-angle glaucoma?	Treatment consists of medications that produces miosis and stretching of the trabeculae. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 4.*
How does mannitol affect intraocular pressure?	Mannitol, dextran, urea, and sorbitol all decrease intraocular pressure by exerting a hyperosmotic effect that decreases aqueous humor formation by the ciliary body. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1376.*
Is atropine contraindicated in patients with glaucoma?	The amount of atropine typically administered preoperatively (0.4 mg IV) results in approximately 0.0001 mg absorbed by the eye which many practitioners regard as having an insignificant effect on intraocular pressure. Scopolamine has a greater mydriatic effect than atropine and is not recommended for use in patients with glaucoma. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1377.*

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What specific actions should you incorporate into the anesthetic for patients with glaucoma? What factors will increase intraocular pressure?

Preoperatively, instill miotic drops to enhance aqueous humor drainage, avoid venous congestion and overhydration, avoid coughing (which increases venous pressure), and avoid hypotension that could promote thrombosis in retinal vessels. Succinylcholine can increase intraocular pressure (IOP). Nondepolarizing agents either decrease the IOP or have no effect on it. Volatile anesthetics, nitrous oxide, opioids, benzodiazepines, and barbiturates all decrease the IOP.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 760-761.

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